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Breast cancer – Spread and prevention

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Spread

Breast cancer initially grows as a primary tumor in the breast (most common location: upper outer quadrant, figure 1). In addition to local growth in size, a malignant tumor may also metastasize; typically, with breast cancer, this is initially lymphogenic, and also hematogenous at a later point.

In patients with a clinical cancer diagnosis (≤ 5 cm, self-examination or during a medical examination), central/retroareolar tumors are significantly larger than those located peripherally in the quadrants (25 vs. 20–21 mm; tumor size in the peripheral regions only differs minimally). [P11]

The following prognosis factors are associated with an elevated risk of metastasis:

- Number of lymph nodes affected by metastasis (TNM classification: pN category)
- Tumor size (pT category)
- Level of tumor differentiation (grading)
- Presence of vascular or lymphatic invasion
- Tumor biology factor (hormone receptor status, Her2 overexpression, Ki-67)

Lymphogenic metastasis

The breast lymph flows from the subepithelial to the subdermal lymphatic vessels and channels into a subareolar plexus. From there, vessels lead deep into the parenchyma and to the axillary lymph nodes, and less frequently to the intercostal and parasternal lymph nodes (internal mammary chain). In rare cases, the "usual" path to the axillary lymph nodes is bypassed, meaning that skip metastases can be found in the high axillary or infra/supraclavicular lymph nodes. Cutaneous lymphatic vessels can also lead to the skin over the contralateral breast or even the contralateral axilla.

As lymph primarily drains into the axilla, this is where lymph node metastases are most frequently found. Only approx. 3 % flows medially to the internal mammary chain, mostly from the deep and thoracic wall-adjacent areas of the breast. In clinical/morphological terms, three levels of axillary lymph nodes are identified (figure 1):

- Level I is lateral to the lateral edge of the pectoralis minor muscle
- Level II is located under the pectoralis minor muscle

- Level III is medial to the medial edge of the pectoralis minor muscle (to the clavicle)

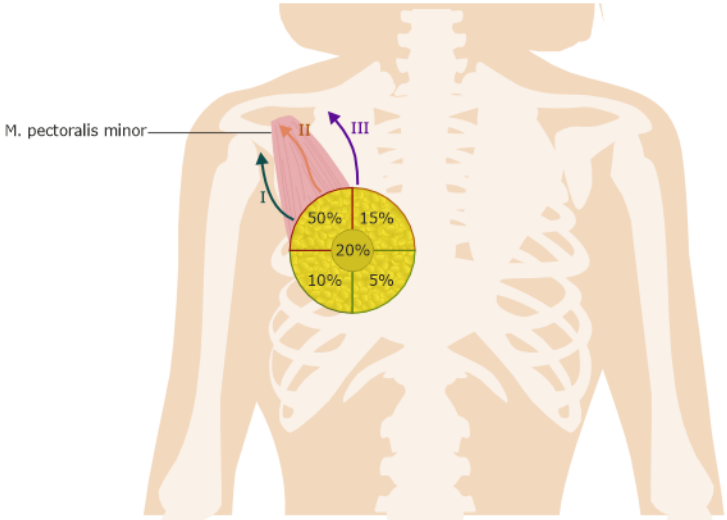

Figure 1: Lymph drainage pathways in breast cancer.

The dermal and parenchymal lymphatic vessels primarily lead to a specific lymph node (or a group of 2 to 3 lymph nodes, but very rarely more). These represent the lymph drainage of the breast and will be the first to be affected in the case of metastasis. These sentinel lymph nodes can be marked, identified during surgery and then removed selectively (figure 2). If these sentinel lymph nodes are tumor-free, it can then be assumed that all other axillary lymph nodes are also tumor-free. Since the method was introduced in the early 2000s, it has quickly become a standard procedure and just a few years later, two-thirds of patients with breast cancer who underwent surgery were treated with this tissue-preserving approach. The "traditional" lymphadenectomy has therefore been replaced as the standard procedure for surgical staging of the axilla. [B11][B23]

Peritumoral or subareolar radioactive colloid injection, usually technetium-99m. The tracer is transported to the regional sentinel lymph node (usually in the axilla) via the lymphatic vessels. The radioactively marked lymph node can then be identified during surgery and removed. In the early part of the “sentinel age”, blue dye was also used for marking; this has been largely abandoned in recent years. [B21]

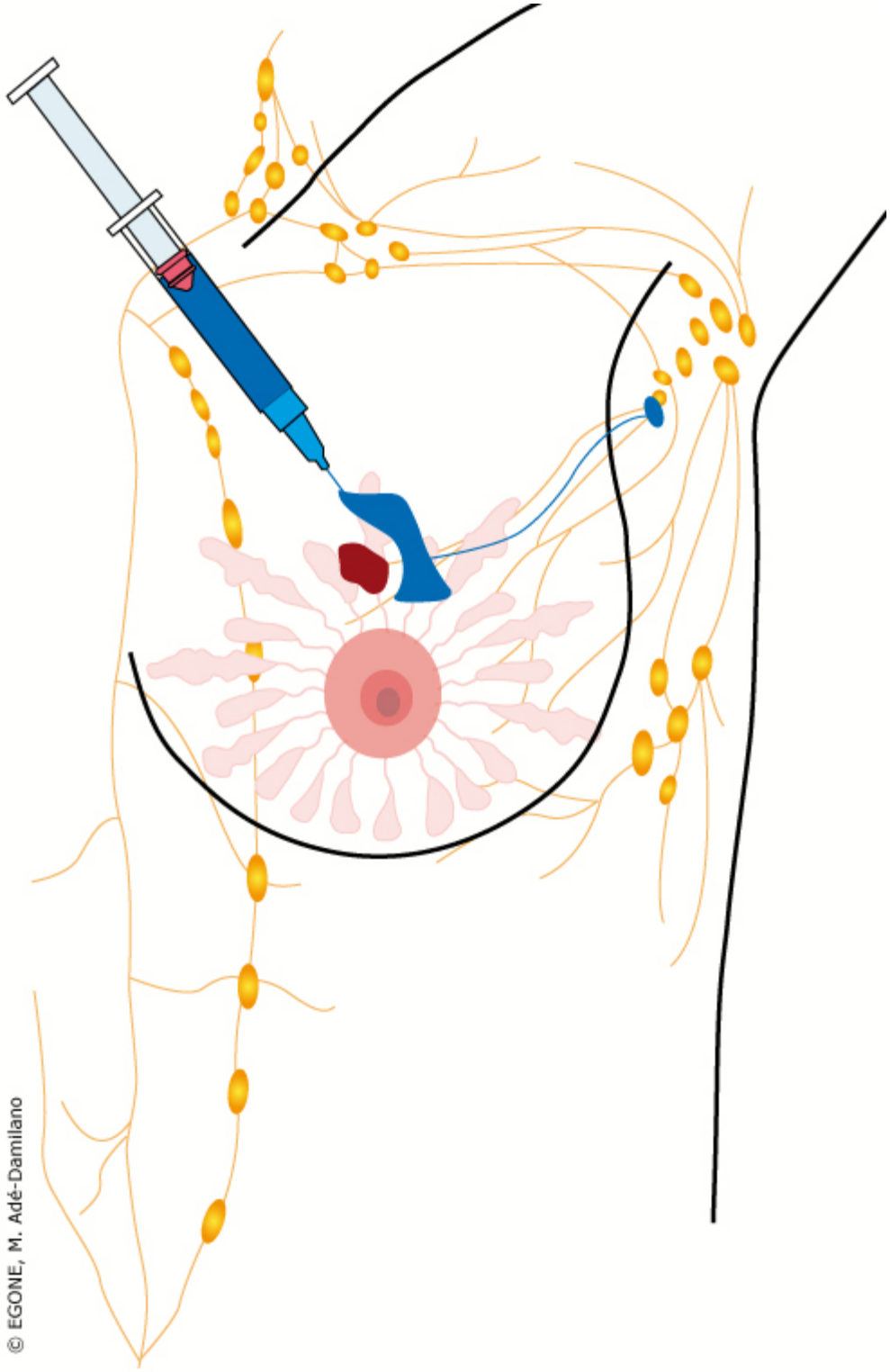

Figure 2: Principle of sentinel lymph node marking.

Hematogenous metastasis

Hematogenous metastasis (distant metastasis) has a much greater bearing on a patient's prognosis in terms of disease-specific survival than locoregional lymph node metastasis. If distant metastasis is present, current therapies can only stabilize and delay the disease (potentially for several years with hormone-receptor positive cancers). However, recovery is only possible in rare cases [B13]. Distant metastasis may be present when the breast cancer is first diagnosed (primary metastasis, approximately 6 % of all newly diagnosed breast cancers). However, secondary metastasis, where metastases occur after a disease-free period potentially lasting several years following completion of primary treatment, is more common. [B13]

Distant metastasis of the following organs is typical on first diagnosis (increasing frequency):

- Skeleton with osteolytic and/or osteoplastic metastases (62 %): spine, proximal femur, pelvis, ribs
- Lungs/pleura (38 %)
- Liver (26 %)
- Brain (7 %)
- Non-regional lymph nodes (25 %)
- Other organs (10 %)
  [B12][B22]

Prevention

Primary prophylaxis

In principle, primary prevention can be achieved by:

- Reducing risk factors
- Taking risk-reducing substances
- Risk-reducing surgical measures

Reducing risk factors

Unfortunately, nothing can be done to tackle the two most important risk factors for breast cancer, namely “Age” (lifetime risk 1 in 8; average age on first diagnosis in Switzerland: 61-62 years) and “female sex” (risk approx. 100 times higher than for men).

The cancer prevention guidelines published by the World Cancer Research Fund (WCRF) and the American Cancer Society (ACS) recommend that women pay careful attention to maintaining a healthy weight, exercising regularly (approx. 150 minutes per week, moderate intensity), eating a balanced diet low in meat and high in plant nutrients, and drinking only limited quantities of alcohol over the long-term [B02, B24]. A long-term lifestyle with regular exercise and limited alcohol consumption in particular reduces the risk of breast cancer. Avoiding excess weight prevents breast cancer in postmenopausal women. It is unclear whether a healthy diet has a positive impact on the risk of breast cancer, although it certainly plays an important role in preventing other common diseases such as cardiovascular diseases and diabetes. As the developing breast is also exposed to factors affecting the risk of later breast cancer (for example smoking), preventive measures should also be implemented by adolescents and young adults. Some estimates suggest that long-term lifestyle changes could prevent up to a quarter of breast cancer cases [B14]. [B02][B14][B24]

Epidemiological studies have demonstrated that parity can affect the risk of developing cancer. A lower risk is associated with

- Early first birth (< 30 years) and multiparity (risk reduction inversely proportional).
- Breastfeeding for 6 months.
  [B04]

Above all, the recommended measures for primary prevention make it clear that there is no specific prophylaxis in terms of general lifestyle (as is clear, for example, with avoiding smoking and the prevention of bronchial cancer). Recommendations are either very general (healthy diet, exercise, avoiding alcohol and smoking, etc. are positive factors for preventing almost all diseases) or are only useful as a hypothesis on carcinogenesis (the protective effect of early motherhood with breastfeeding is based on the influence of hormonal factors in the development of breast cancer - but no woman has and breastfeeds a baby at 25 years old simply to avoid getting breast cancer as a 55 year old).

- A meta-analysis showed a relative risk reduction of 4,3 % per 12 months breastfeeding and, in addition, a relative risk reduction of 7 % for each birth [B04]
- For postmenopausal breast cancer, there is consistent epidemiological evidence and a clear dose-response curve that shows that a higher proportion of body fat increases risk [B02]
- For premenopausal breast cancer, the opposite applies: A body mass index (BMI) ≥ 31 kg/m2 reduces the risk of breast cancer by 46 % versus a BMI 21 kg/m2 [B02]
- Interestingly, Swiss data has shown that a cohort of postmenopausal breast cancer patients did not have higher rates of overweight or obesity than an age-appropriate representative Swiss normal cohort. We have therefore not been able to reproduce the positive association between body weight and breast cancer development found in populations with higher rates of overweight/obese women in Switzerland, a country with a lower rate of overweight/obese women. This may be because there is an exponential correlation between BMI and postmenopausal breast cancer. This would mean that a positive correlation between overweight/obesity and breast cancer development is primarily found in populations in countries with a high proportion of obese and morbidly obese women. With regard to the prevalence of obesity, Switzerland has been at the lower end of the European league table for many years. It is also possible that the correlation between BMI and breast cancer risk found in some countries with high rates of obesity, such as the United States, is not associated with BMI alone, but a reflection of a social environment and lifestyle that involves many difficult to control factors and leads to the majority of people being overweight or obese [P06].
- Long-term combined hormone replacement therapy (>3 to 5 years estrogens and progestogens) is a risk factor [P01].

- Physical activity (exercise) reduces post-menopausal breast cancer by approx. 25 %. Exercise can delay the onset of menstruation in young women and thus reduce the risk [P09].
  [B02][B04][P01][P06][P09]

Risk-reducing drugs (chemoprevention)

Four randomized studies have shown that chemoprevention with selective estrogen receptor modulators (SERMs) for five years reduces the incidence of invasive breast cancer in high-risk patients (family history, age, LCIS, atypical hyperplasia) by approx. 30 to 50 %. Tamoxifen and raloxifene are the main drugs that have been studied. They are both partial estrogen antagonists that inhibit epithelial cell proliferation in the breast gland tissue. However, they act as estrogen agonists on bone (leading to a significant reduction in the risk of fractures), the liver and the coagulation system. Unlike raloxifene, tamoxifen also has a stimulating effect on the endometrium. [B03][B19]

The NSABP (National Surgical Adjuvant Breast and Bowel Project) Breast Cancer Prevention Trial (P-1) showed that tamoxifen 20 mg/d for 5 years reduced the risk of invasive cancer by 43% and non-invasive cancer by 37%. However, this reduction only applied to estrogen receptor-positive tumors, not those that were estrogen receptor-negative. Tamoxifen was effective in all age groups, with a tendency to be more effective in women over the age of 60. This group also experienced more side effects (see below) [P12]. Another study (International Breast Cancer Intervention Study I: IBIS-I study) showed a significant risk reduction for invasive cancer of 0,73 on administration of 20 mg tamoxifen versus placebo for 5 years. After 10 years, the HR was 0.69; this effect may also be attributable to the reduction in gland tissue density in older women. Similarly to other studies, in this study only the development of estrogen receptor-positive tumors was prevented [P02]. Raloxifene (Evista®) also reduces the risk of developing an invasive hormone receptor-positive cancer by 56% [B19]. In a direct comparison of tamoxifen 20 mg/d and raloxifene 60 mg/d (STAR Study of Tamoxifen and Raloxifene), tamoxifen was significantly more effective than raloxifene in preventing invasive cancer; however, this was at the expense of an increased risk of developing endometrial cancer and thromboembolic events [P12]. There is no reliable data of the prevention of male breast cancer. [B19][P02][P12]

Undesired adverse events are predominantly thromboembolic events. The relative risk of pulmonary embolism under tamoxifen was 2.15; the risk of thromboembolic events under raloxifene was 1.6. As tamoxifen, unlike raloxifene, acts as an estrogen agonist on the endometrium, the incidence of endometrial was significantly increased in women aged over 50 years (RR: 3.28). However, hot flashes and leg cramps were more common with raloxifene. [B08]

Two studies have shown that aromatase inhibitors (anastrozole: Arimidex®, generics; exemestane: Aromasin®, generics) have a preventive effect on both invasive and preinvasive hormone receptor-positive breast cancer: In the International Breast Cancer Intervention Study II (IBIS-II study) the risk under anastrozole fell significantly by 53 % versus placebo; in the Mammary Prevention Trial 3 (MAP-III study) the risk reduction under exemestane was 65 %. The treatment-related adverse effects included hot flashes, musculoskeletal symptoms and accelerated bone loss [P03, P05]. The data is less convincing for women who carry the BRCA1/2 genes. However, smaller studies show good indications that tamoxifen reduces the risk of breast cancer at least in women with the BRCA2 gene [P08]. Nevertheless, risk-reducing mastectomy is more effective in this population (see below). [P03][P05][P08]

According to the ASCO guidelines, endocrine prevention with tamoxifen and/or raloxifene for five years should be considered for women with an increased risk of breast cancer and discussed with the patient. However, as an improvement in breast cancer-specific or overall survival has not been proven for either substance, this recommendation is controversial:

- Proponents of chemoprevention see every prevented case as an argument for drug-based intervention.
- Critics justify their position with the fact that chemoprevention may not actually improve breast cancer mortality and the (in their opinion too few) benefits do not outweigh the potential side effects and complications of endocrine therapy over several years.

The use of low-dose medication with tamoxifen 5 mg/d is an interesting new development in this respect. Here, it has been shown that even with this low dose, taken over just three years, a significant reduction in new neoplastic diseases of the breast could be achieved in an at-risk population (500 patients with biopsy-confirmed atypical ductal hyperplasia and lobular or ductal carcinoma in situ) (HR: 0,48), with only limited treatment-related adverse effects. [B05]

Risk reduction through surgical ablation

For approx. 5 to 6 % of breast cancer patients, the disease can be traced back to a genetic mutation, particularly of the BRCA1 and BRCA2 genes. Women with these mutations have a much higher risk of developing both breast cancer (BRCA1: 50-85 %; BRCA2: 33-54 %) and ovarian cancer (BRCA1: 22-51 %; BRCA2: 4-21 %) [B17]. Bilateral risk-reducing mastectomy can reduce the incidence of breast cancer in these patients by more than 90 % [B06, B15, B20]. The younger the woman when the procedure takes place, the stronger the breast cancer-preventing effect [B10]. Surgical options with regard to how radical the mastectomy is (“traditional” mastectomy; skin-sparing mastectomy, i.e. removal of the gland parenchyma but preservation of the skin; nipple-sparing mastectomy, i.e. removal of the gland parenchyma but preservation of the skin and nipple-areolar complex) must be discussed with the patient. Today, most women choose to have primary breast reconstruction. In carriers of the BRCA1/2 mutation, risk-reducing bilateral adnexectomy alone can reduce the incidence of primary breast cancer by 50 %, and the reduction in the risk of fallopian tube/ovarian cancer is even higher (80 %) [B16]. In a prospective study of BRCA1/2-associated breast cancer, bilateral adnexectomy also showed a positive effect on overall and disease-specific survival [B07]. [B06][B07][B15][B10][B16][B17][B20]

Secondary prophylaxis

The most important method of secondary prophylaxis is breast cancer screening by means of mammography (figure 3). This is an x-ray examination of the mammary glands using two views, which can be used for a truly early diagnosis of breast cancer. In general, breast cancers which are detected early have a better prognosis and their treatment is often simpler and less aggressive than diseases in an advanced stage.

Screening vs. early detection: In order to avoid giving patients false expectations with regard to the potential of the method, it is important to use the right terminology. A “screening” mammography is not a “preventive” examination, but an examination with the objective of early detection. A comparison with cervical screening (a “smear” test) makes this clear: during cervical screening, a cytological smear is used to identify premalignant changes to the cervix (dysplasia: CIN I-III), which could, sometimes over many years, develop into cancer. From a certain stage of development, a surgical procedure can then be carried out. Screening means that the precancerous growth is identified early and stopped, so that the cancer does not actually develop. With mammography, this form of screening is almost impossible, although in isolated cases, non-invasive (or not-yet-invasive) changes can be identified (e.g. ductal carcinoma in situ [DCIS], which presents at a subclinical stage with typical grouped polymorphic microcalcifications). The actual objective of mammography is to identify small invasive tumors at a size that means that they are not yet palpable. Generally, mammography does not show the cancer in its early non-invasive form (“screening”), but with the lowest possible level of morphological spread (“early detection”). The idea behind this being that we don’t want to only find a cancer when it is palpable for the patient (e.g. 2.5 cm), but rather when it is still a lesion that is not palpable but can only be seen with mammography (e.g. 7 mm). Ideally, the difference in detected tumor size (7 mm vs. 2.5 cm) will be the difference whether or not the patient can be cured.

Regular self-examination of the breasts is a useful personal screening method, not least as it improves body awareness. However, studies have not been able to show that this measure contributes to a reduction in breast cancer mortality. Conversely, it leads to increased diagnostic activity and more frequent diagnosis of benign diseases versus patients who do not carry out self-examination. [P07]

A “screening” mammography is a population-based routine examination for asymptomatic women: Ideally, screening is provided in a highly formalized national (in Switzerland cantonal) program. If it is indicated due to symptoms or changes affecting the breast (e.g. pain, new palpation findings, skin retraction, etc.), this is a diagnostic mammography. Major randomized studies have shown that screening mammography reduces breast cancer mortality in women aged between 50 and 70 years by 25 to 31 % [B18]. However, it is conceivable that the increasingly successful multimodal treatment of breast cancer will relativize the efficacy of screening mammography in the future. [B18]

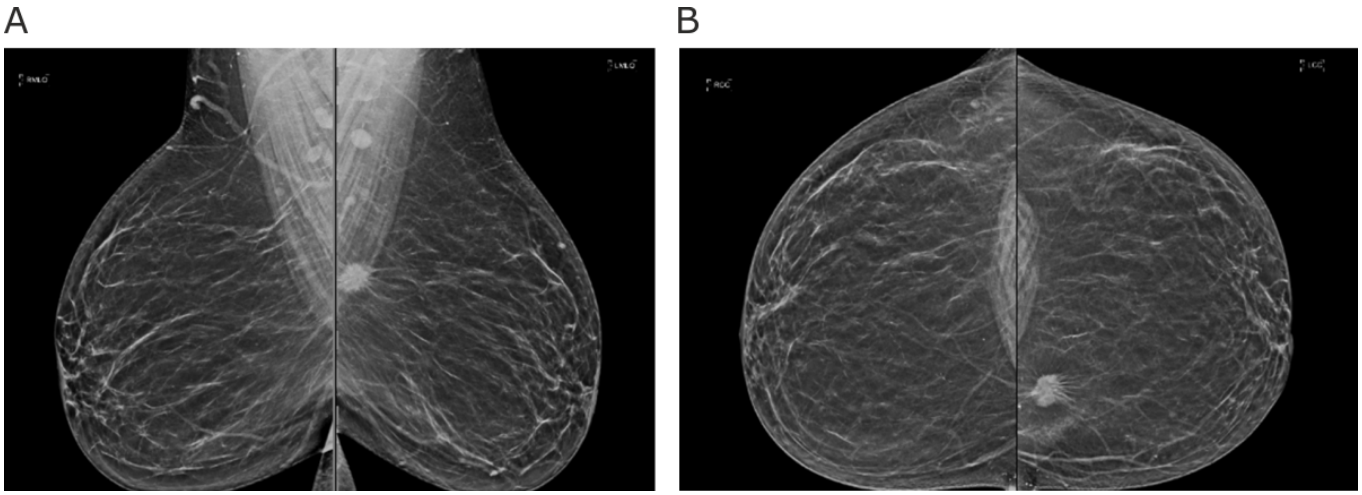

Figure 3: Bilateral mammography (Images provided by Dr S. Dubach, Zurich Breast Center).
A) Mediolateral view, low gland parenchyma density, ACR b
B) Craniocaudal view
Case: 50-year-old postmenopausal patient with: invasive ductal carcinoma on left at 11-o’clock position cT1c (18 mm) cN0 cM0, G2 ER/PR 100%, Her2 negative, Ki-67: 10%

If the mammography shows high gland tissue density (ACR c-d), it is not always possible to identify small tumors reliably. In this case, an additional ultrasound examination is recommended.

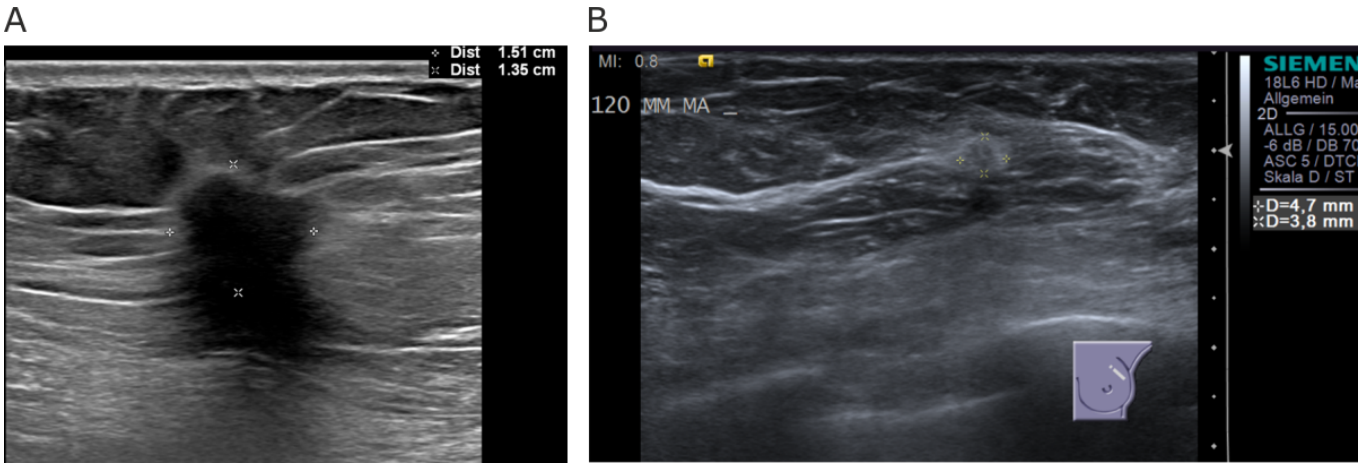

Figure 4: Ultrasound (Images provided by Dr S. Dubach, Zurich Breast Center).
A) Presentation of a tumor measuring 15x13 mm. Case: 46-year-old premenopausal patient with: invasive ductal carcinoma on left cT1c (15 mm) cN0 cM0, G2 ER 100%, PR 80%, Her2 negative, Ki-67: 15%
B) Presentation of a tumor measuring 4x5 mm.
Case: 62-year-old patient with: invasive ductal carcinoma on right at 2-o'clock position cT1a (5 mm) cN0 cM0, G1 ER/PR 100%, Her2 negative, Ki-67: 5%

Breast MRI (magnetic resonance mammography) has higher sensitivity than conventional mammography, but lower specificity [P138]. There is not yet any evidence of a positive effect on overall mortality in the general population. For cost reasons, it is therefore only recommended in women with a very high risk of breast cancer (e.g. carriers of BRCA1/2 mutations) or in addition to mammography in certain cases as a screening method. [P13]

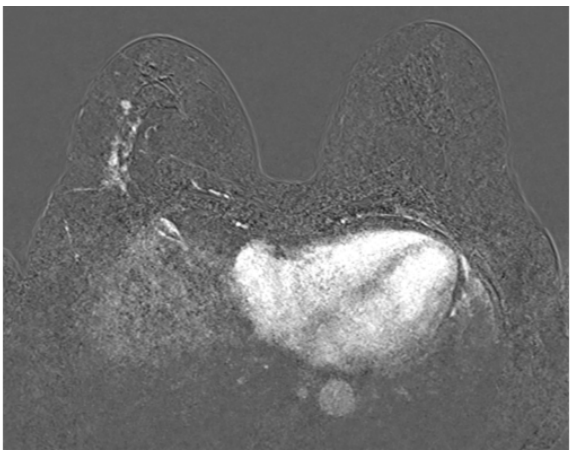

Figure 5: Bilateral breast MRI (Image provided by Dr S. Dubach, Zurich Breast Center). Case: 71-year-old patient with: low-grade DCIS on right at 9-o'clock position pTis (70 mm) cN0 cM0, R0, ER/PR 100%
Clarification: MRI-guided vacuum biopsy (right breast at 9-o'clock position, central and peripheral). Treatment: Segmentectomy on right.

In Switzerland, the obligation of health insurance providers with regard to screening mammography is linked to a cantonal program with precisely defined quality requirements. While mammography screening is already in use in a large part of Europe, this does not apply to all Swiss cantons. The “longest history” of mammography screening is found in the French-speaking cantons: the first screening program was established in the Cantons of Vaud, Geneva and Valais in 1999; the first German-speaking cantons (Thurgau, St. Gallen and Graubünden) only began their program in 2010/2011.

Cantonal mammography screening program in Switzerland

The introduction of an official mammography screening program clearly increases the willingness of women to undergo a mammography. An evaluation of the 2012 Swiss Health Survey with regards to participation in mammography examinations, 78 % of women aged 50 to 69 years in French-speaking cantons said that they had undergone a mammography in the last two years; in German-speaking cantons this rate was only 35 %. [P04]

In the cantons without a screening program, health insurance providers only cover the examination if there is a medical referral on the basis of a finding. For women aged under 50, the cost/benefit ratio and risk/benefit ratio of screening mammography are lower. For this reason, Swiss health insurance providers are not obligated to cover the cost of the examination for this age group, unlike in other countries. Health insurance companies cover at most the cost of one mammography per year, if there is a family history of breast cancer concerning the patient’s mother, sister or daughter.

In recent years, there has been much critical discussion of mammography screening. In addition to high cost, the following potential disadvantages are also highlighted:

- False-positive results: Suspect findings that lead to further investigation, but no cancer is found.
- False-negative mammography: Cancer that is not identified despite mammography.
- Overdiagnosis: Identification of cancer that has no clinical relevance with regard to the patient’s lifetime.
- Lead-time-bias: Identification of the cancer leads to earlier diagnosis but no extension of overall survival.
- Radiation exposure: As mammography is an x-ray-based examination, it must

be assumed that the ionizing radiation itself could contribute to the induction of breast cancer. Although this theoretical risk can be calculated, it has not been directly proven to date. The risk for women with a known BRCA mutation may be somewhat higher.

- Pain caused by mammography: For some women, the compression of the mammary gland leads to very unpleasant pain, often with negative effects on compliance. Many women assume that compression is damaging, although this has never been proven. (Pressing the mammary gland between two plates distributes the gland volumes better on a surface, meaning that the diameter is lower. This leads to better image quality and lower radiation exposure).
- Many of these considerations are due to problems that are inherent to the system, and in principle apply to all diagnostic aspects of modern medicine.

Literature

Nr.	Title	Author	Journal	Volume	Pages
[B01]	Research & Guidelines. Breast cancer	American Society of Clinical Oncology (ASCO)			
[B02]	World Cancer Research Fund International: Continuous Update Project-systematic literature review and meta-analysis of observational cohort studies on physical activity, sedentary behavior, adiposity, and weight change and breast cancer risk	Chan DSM, Abar L, Cariolou M, Nanu N, Greenwood DC, Bandera E, McTiernan A, Norat T	Cancer Causes & Control	30	1183-120(
[B03]	Breast Cancer Chemoprevention: Targeting the Estrogen Receptor	Chlebowski RT, Prabhaker R, Tomoko Tagawa T	American Journal of Hematology/ Oncology	12	6-13

Nr.	Title	Author	Journal	Volume	Pages
[B04]	Breast cancer and breastfeeding: collaborative reanalysis of individual data from 47 epidemiological studies in 30 countries, including 50302 women with breast cancer and 96973 women without the disease	Collaborative Group on Hormonal Factors in Breast Cancer	Lancet	360	187-195
[B05]	Randomized Placebo Controlled Trial of Low-Dose Tamoxifen to Prevent Local and Contralateral Recurrence in Breast Intraepithelial Neoplasia	DeCensi A, Puntoni M, Guerrieri-Gonzage A, Caviglia S, Avino F, Cortesi L, et al.	J Clin Oncol	19	1629-1637
[B06]	Association of Risk-Reducing Surgery in BRCA1 or BRCA2 Mutation Carriers with Cancer Risk and Mortality	Domchek SM, Friebel TM, Singer CF, Evans DG, Lynch HT, Isaacs C, et al.	JAMA	304	967-975

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[B07]	Breast cancer risk following bilateral oophorectomy in BRCA1 and BRCA2 mutation carriers: an international case-control study	Eisen A, Lubinski J, Klijn J, Moller P, Lynch HT, Offitt K, et al	J Clin Oncol	23	7491-7496
[B08]	Benefit/risk assessment for breast cancer chemoprevention with raloxifene or tamoxifen for women age 50 years or older.	Freedman AN, Yu B, Gail MH, Costantino JP, Graubard BI, Vogel VG, et al.	J Clin Oncol	29	2327-2333
[B09]	Diseases of the breast. 5th edition	Harris JR, Lippmann ME, Morrow M, Osborne CK (Eds)	Lippincott, Williams & Wilkins		
[B10]	The expected benefit of preventive mastectomy on breast cancer incidence and mortality in BRCA mutation carriers, by age at mastectomy	Giannakeas, V., Narod, S.A.	Breast Cancer Res Treat	167	263–267

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[B12]	Primary and secondary distant metastatic breast cancer: two sides of the same coin.	Güth U, Magaton I, Huang DJ, Fisher R, Schötzau A, Vetter M	Breast	23	26-32
[B13]	Long-term survival and cure in distant metastatic breast cancer	Güth U, Elfgen C, Montagna G, Schmid SM	Oncology	97	82-93
[B14]	Can Diet and Lifestyle Prevent Breast Cancer: What Is the Evidence?	Harvie M, Howell A, Evans DG	ASCO Educational Book e67		
[B15]	Risk-reducing surgery increases survival in BRCA 1/2 mutation carriers unaffected at time of family referral	Ingham SL, Sperrin M, Baildam A, Ross GL, Clayton R, Lalloo F, et al.	Breast Cancer Res Treat	142	611-618

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[B17]	Breast cancer risks in women with a family history of breast or ovarian cancer who have tested negative for a BRCA1 or BRCA2 mutation	Metcalfe KA, Finch A, Poll A, Horsman D, Kim-Sing C, Scott J, et al	Br J Cancer	100	421-425
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[B21]	Sentinel lymph node biopsy in primary breast cancer: Trust the radiolabeled colloid method and avoid unnecessary procedures	Schmid SM, Myrick ME, Forrer F, Obermann EC, Viehl CT, Rochlitz C, Güth U	Eur J Surg Oncol	37	211-216
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Nr.	Title	Author	Journal	Volume	Pages	Year
[P05]	Exemestane for breast-cancer prevention in postmenopausal women	Goss PE, Ingle JN, Ales-Martinez JE, Cheung AM, Chlebowski RT, Wactawski-Wende J, et al	N Engl J Med	364	2381-2391	2011
[P06]	Entwicklung von Übergewicht und Adipositas beim Mammakarzinom und ihre Bedeutung für Diagnose und klinisches Management	Güth U, Schmid S, Eichholzer M	Schweiz Rundsch Medizin Prax	102	473-480	2013
[P07]	Breast self-examination and death from breast cancer: a meta-analysis	Hackshaw AK, Paul EA	Br J Cancer	88	1047-1053	2003

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[P08]	Tamoxifen and breast cancer incidence among women with inherited mutations in BRCA 1 and BRCA 2: National Surgical Adjuvant Breast and Bowel Project (NSABP-1) Breast Cancer Prevention Trial	King MC, Wieand S, Hale K, Lee M, Walsh T, Owens K, et al.	JAMA	286	2251-2256	2001
[P09]	Physical activity and breast cancer prevention	Lynch BM, Neilson HK, Friedenreich CM	Recent Results Cancer Res	186	13-42	2011
[P10]	Total skin-sparing mastectomy in BRCA mutation carriers	Peled AW, Irwin CS, Hwang ES, Ewig CA, Alvarado M, Esserman LJ	Ann Surg Oncol	21	37-41	2014

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[P11]	Self-detection and clinical breast examination: comparison of the two “classical” physical examination methods for the diagnosis of breast cancer.	Schwab FD, Huang DJ, Schmid SM, Schötzau A, Güth U	Breast	24	90-92	2015
[P12]	Update of the National Surgical Adjuvant Breast and Bowel Project Study of Tamoxifen and Raloxifene (STAR) P-2 Trial: Preventing breast cancer	Vogel VG, Costantino JP, Wickerham DL, Cronin WM, Cecchini RS, Atkins JN	Cancer Prev Res (Phila)	3	696-706	2010
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